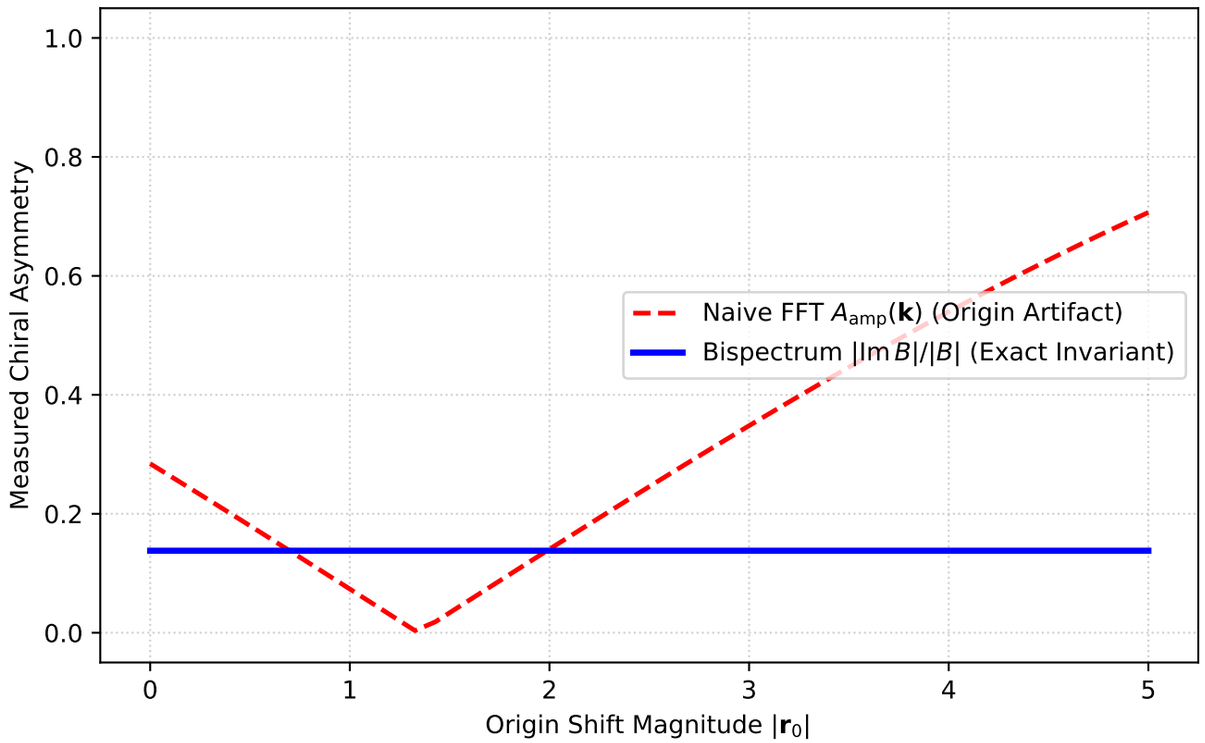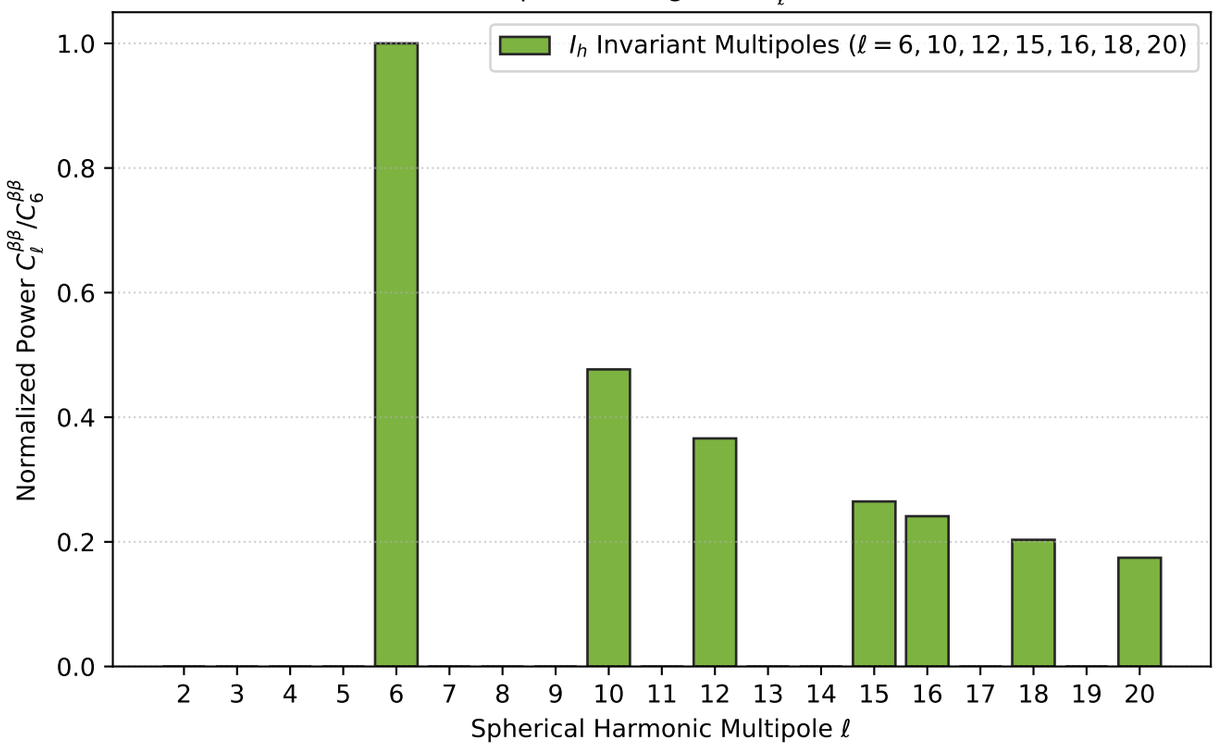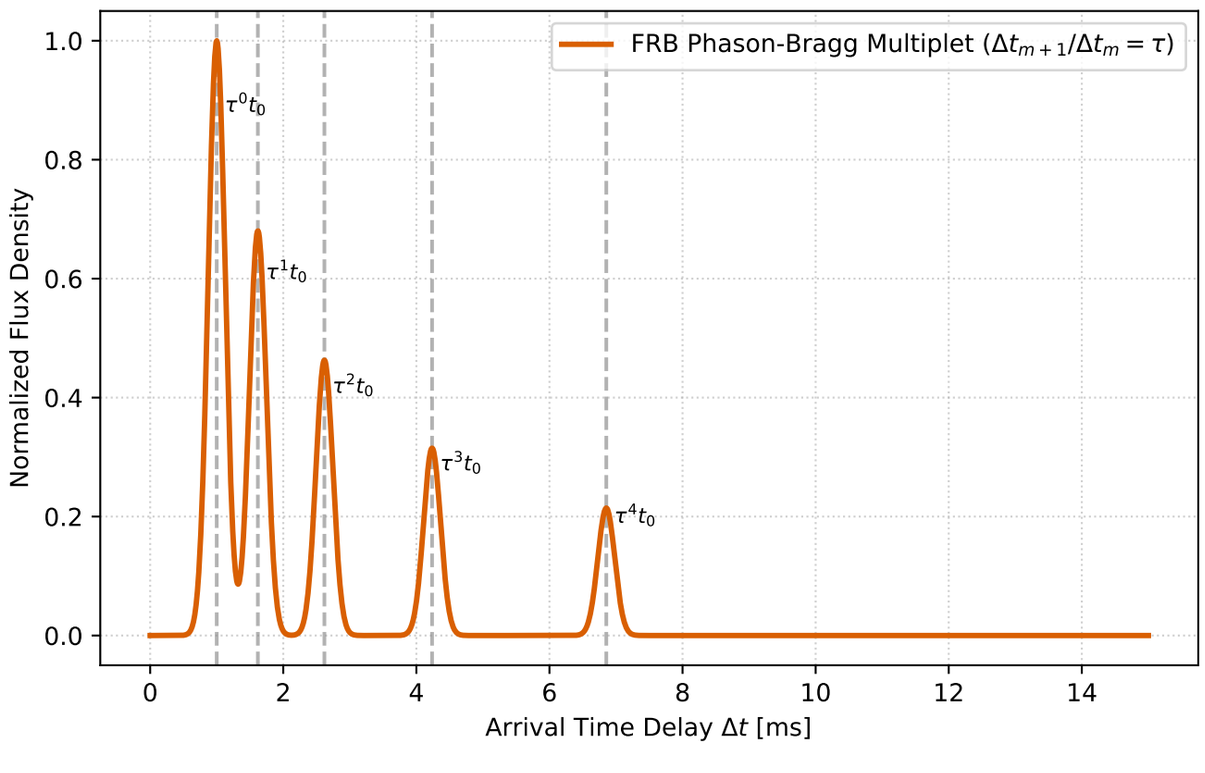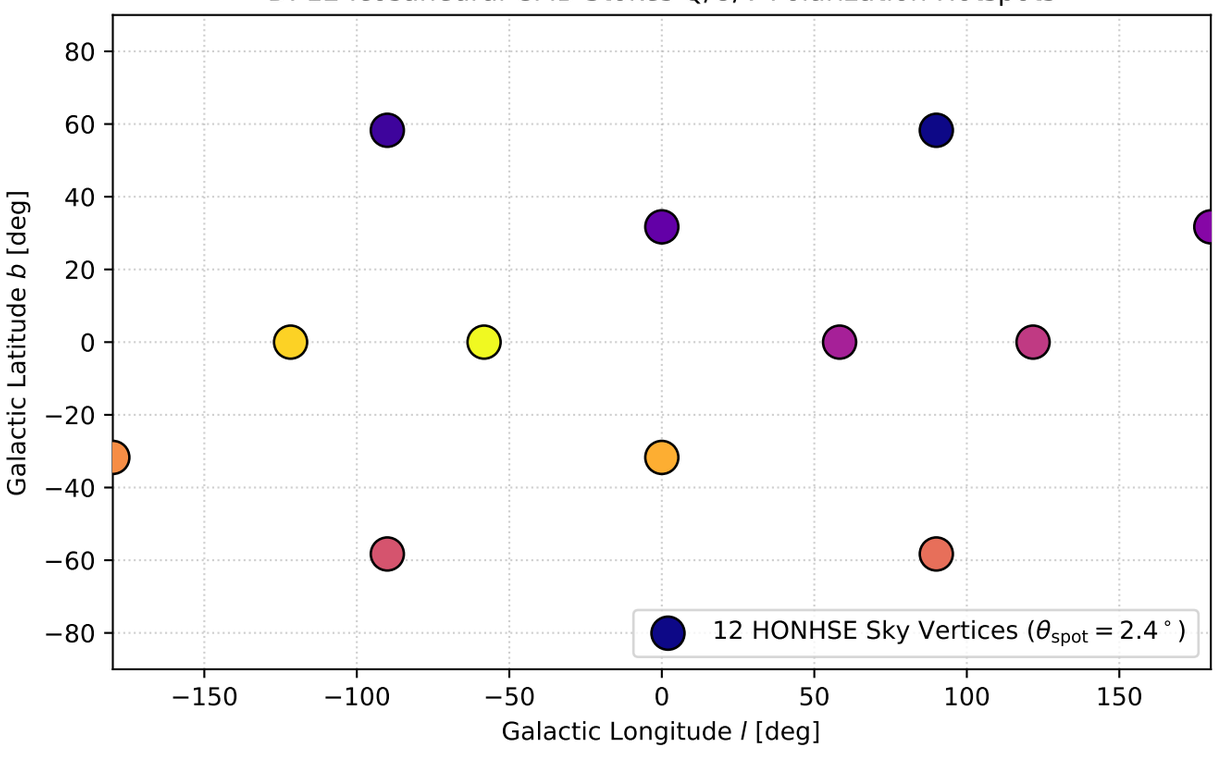Pillar 9: Aperiodic Substitution Cosmology, Bispectrum Audit & Multi-Messenger Signatures